

Investigating the Neural Markers for Attention Hold

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NEURAL MARKERS OF ATTENTION HOLD

Abbreviation	Definition
RT	Response Time / Reaction Time
ISI	Interstimulus Interval
fMRI	Functional Magnetic Resonance Imaging
ERP	Event-Related Potentials
LPFC	Left Lateral Prefrontal Cortex
MFG	Middle Frontal Gyrus
IPS	Intraparietal Sulcus
TPJ	Temporoparietal Junction
ABV	Attention Bias Variability
SOA	Stimulus Onset Asynchrony
EEG	Electroencephalography
SPCN	Sustained Posterior Contralateral Negativity
CDA	Contralateral Delay Activity
CNSW	Contralateral Negative Slow Wave
CSA	Contralateral Search Activity
MEG	Magnetoencephalography
LPC	Late Positive Component
IIN	Ipsilateral Invalid Negativity
FFA	Fusiform Face Area
P_D	Distractor Positivity
ANOVA	Analysis of Variance
EOG	Electrooculogram
CFB	Chicago Face Database
CI	Confidence Interval

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Introduction

Our visual systems are exposed to many sensory inputs at any given time (Hickey et al., 2006; Theeuwes & Godijn, 2002); however, human cognitive processes have a finite capacity. As a result, we rely heavily on attentional mechanisms to prioritize important information (Hickey et al., 2006; Hopfinger & Parks, 2012; Rauschenberger, 2003; Ruz & Lupiafiez, 2002). We choose the information that is relevant to our objectives, helping us in the higher-order cognitive processes, such as reasoning, critical decision making, and memory storage (Anderson, 2013; Hickey, et al., 2006; Theeuwes, 1994). This emphasizes the significance of attentional mechanisms, which we aim to investigate in this study through neuroimaging techniques to uncover specific components associated with these mechanisms.

Attention Capture

Some particularly salient stimuli have the intrinsic capacity to draw our attention, regardless of how relevant they are to our immediate aims (Hickey et al., 2006; Yantis, 2000). For example, when a flash of light, a vivid color, or a moving item appears, our attention is involuntarily captured (Folk, Remington, & Wright, 1994; Hopfinger & Parks, 2012). Existing research proposes that attention is regulated by two separate modes: endogenous and exogenous. The endogenous mode (also known as goal-driven mechanism, top-down or goal-driven control of attention), refers to the intentional directing of attention. Alternatively, exogenous mode (also known as salience-driven mechanism, bottom-up or stimulus-driven control of attention) is the involuntary capture of attention by new stimuli or changes in stimuli (Akimoto et al., 2013;

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Anderson, 2013; Folk, Remington, & Johnston, 1992; Lien et al., 2010; Theeuwes, 1991a, 1991b, 1992, 2004, 2010). The potential of stimuli to generate a completely stimulus-driven orienting of attention is referred to as attentional capture (Hickey et al., 2006; Sawaki, & Luck, 2010; Theeuwes, 1991a). The term involuntary attention is used to describe the capturing of our attention to the stimuli that we are unable to stop attending to (Anderson, 2013; Hopfinger & Parks, 2012), independent of any top-down attentional control settings (Lien et al., 2010; Rashal et al., 2022).

One way to measure the capture of attention is by using a spatial cuing paradigm, specifically a pre-cueing task, as used by Ruthruff et al. (2020). In this task, participants are presented with a cue (a color singleton or an abrupt onset) at a random location in the search display that is nonpredictive of the upcoming target location. The critical measure of capture is the cue validity effect, which refers to the degree to which response time (RT) is slower when the pre-cue is invalid (i.e., not at the target location) compared to when it is valid (at the target location). This allowed the researchers to assess the impact of attentional capture on search performance.

Attention Hold

As mentioned earlier, some salient stimuli can capture our attention. However, the costs and benefits of attention capture depend largely on how long attention dwells on the nontarget items during visual search (Ruthruff et al., 2020). Therefore, there is a distinction between attention capture and attention holding as separable cognitive processes: capture of attention relates to how stimulus characteristics elicit attention to the stimulus and draw fixation towards it, while holding of attention refers to how stimulus characteristics keep attention for some length of time on the stimulus (Richards, 2010). In other words, attention hold refers to the sustained focus and maintenance of attention on a specific stimulus or task over a period of time (Parks &

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Hopfinger, 2008). Once attention has been captured or directed towards a particular item, concept, or event, attention hold involves keeping that focus engaged and preventing it from shifting to other stimuli. This hold of attention allows individuals to process and interact with the stimulus in more depth, enabling further cognitive processing, analysis, and response (Hopfinger & Parks, 2012).

Moreover, attention holding may be more essential than initial orientation in influencing cognition and future behaviors (Hopfinger & Parks, 2012; Parks & Hopfinger, 2008). Maintaining attention to the tasks that we do on a daily basis is almost always important for successful performance (Rosenberg et al., 2015). Although the reflexive orienting toward salient stimuli is a strong mechanism, the effect of this reflexive orienting may be minimal and transient in the absence of any hold of attention; nevertheless, if attention is held, the capture is complete (Hopfinger & Parks, 2012). Without any attention hold, the impact of reflexive orienting might be minimal and short-lived.

The process of holding of attention contains three functions: (1) the attentional dwell, or time during which additional (potentially higher order) processing of the attended item occurs; (2) the disengagement of resources from the attended item; and (3) the reorienting of attention away from the item (Ozaki & Ogawa, 2009). Although, the initial orienting toward salient stimuli is automatic, that conformity with top-down settings (i.e. cognitive factors, such as task requirements and expectations, that guide and influence attentional processes) influences how long attention remains on that stimulus, with attentional hold increasing for task-relevant distractors (Theeuwes, Atchley, & Kramer, 2000). Therefore, attentional capture is not solely dependent on bottom-up attributes, such as salient physical stimuli. Instead, it is also contingent upon top-down control

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settings, which are associated with mental representations that individuals hold in their minds to accomplish specific tasks (Folk, et al., 1992; 1994).

Different studies on hold of attention have used different measures to assess this aspect of human cognition. For example, one study used heart rate in response to visual stimuli as a measure of holding of attention in infants, with decreased speed of heart rate after the initial attention orientation measuring attention holding (Richards, 2010). In other studies, using behavioral paradigms, longer reaction times to the target when a distractor cue was present (either simultaneously or sequentially with different inter-stimulus intervals) were used as a measure of holding attention (i.e. attentional dwell time) (Ozaki & Ogawa, 2009; Parks, et al, 2014). That is the case for both task-relevant and task-irrelevant distractors. For example, in Ozaki and Ogawa's study (2009), attention hold was quantified through the assessment of attention dwell time devoted to task-relevant queries. This study measured attention hold by presenting participants with visual cues and Gabor patch stimuli, which required them to maintain focus on a specific location. Participants, inside an MRI scanner, viewed two colored (green and red) Gabor patches positioned in the upper left and right quadrants of their visual field. Each trial began with a fixation cross, followed 4000 ms later by a spatial or neutral cue. The spatial cue directed participants' attention to a specific Gabor patch (left or right), while the neutral cue directed their attention to the fixation cross. After the cue disappeared, participants were instructed to hold their attention on the designated location. After a variable delay (6000 or 8000 ms), one Gabor patch changed color, prompting participants to press a button as quickly as possible in response. Trials were classified into four conditions: valid (cued patch changed color), invalid (un-cued patch changed color), neutral (attention held on the fixation cross), and empty (no cue given). Reaction times (RTs) across these conditions allowed researchers to measure attention hold by calculating costs (slower

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RTs in invalid trials compared to neutral) and benefits (faster RTs in valid trials compared to neutral), reflecting the impact of attentional focus on response speed and attentional allocation.

Conversely, in Parks et al.'s study (2014), the attention hold was measured based on the time spent on the distractor, which comprised visual stimuli such as images of faces and places. In one of the experiments in this study, participants were tasked with maintaining attention on a centrally located fixation point while performing a peripheral discrimination task. In the upper right visual field, a red letter "T" appeared continuously, changing its orientation randomly every second. Participants judged the orientation of the "T" (horizontal/vertical or diagonal) and responded by pressing one of two buttons as quickly and accurately as possible. Throughout each experimental run, distractor images (either fearful faces or places) were presented at the center of the screen for 4 seconds each, and participants were instructed to ignore these task-irrelevant stimuli. Distractors were presented randomly, with a randomized interstimulus interval (ISI) of 3-6 seconds to prevent anticipation. This setup measured attention hold by examining how distractors influenced participants' ability to maintain focus on the peripheral target task, despite the presence of centrally located, attention-grabbing stimuli. By using long distractor durations and familiarizing participants with distractors beforehand, the study minimized initial surprise and transient capture, focusing instead on the sustained effect of distractors on attention hold.

In another study by Orellana-Corrales et al. (2021) to evaluate holding of attention, response times had been used as a measure to indicate the holding of attention. The researchers of that study explained that when targets appear after stimuli linked to oneself (i.e., in a spot previously occupied by a self-associated stimulus), they trigger quicker responses compared to targets following stimuli associated with others. This result suggests that the focus of attention lingers at the location of a self-associated stimulus even after it vanishes, signifying holding of

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attention. Recognizable self-associated stimuli can effectively direct and maintain attention more effortlessly than stimuli associated with others, even leading to attentional capture and prolonged attention hold (Orellana-Corrales et al., 2021).

Some other researchers have used another behavioral design to measure the attention bias and hold using a dot-probe paradigm (for example, Holmes et al., 2009; Song et al. 2022; Torrence et al., 2017). In this paradigm, the participant is presented with two stimuli (e.g., a fearful and a neutral face) on opposite sides of a screen, followed by a probe in one of the two locations. By measuring RTs on valid trials (where the probe appears in the same location as the salient stimulus) versus invalid trials (probe in the opposite location of the salient stimulus), researchers can assess attentional allocation, with faster RTs for valid trials suggesting attentional bias toward the stimulus location. In this paradigm, delayed disengagement would be indicated by slower reaction times compared to baseline trials, suggesting that attention remained on the location of the valid cue longer (Torrence et al., 2017). The dot-probe task's main advantage is its adaptability to different contexts, allowing researchers to study attentional mechanisms on various sets of stimuli (Torrence & Troup, 2018).

Each of these measures that were used for investigating attention hold has its own advantages and disadvantages. For example, reaction times, often used in behavioral paradigms, provide a straightforward metric for attentional dwell time and can capture task-relevant engagement. However, they are susceptible to motor response variability and can be influenced by external factors, such as anticipation and task familiarity (Dang, King, & Inzlicht, 2020; Torrence & Troup, 2018). Due to the limitations of behavioral measures, some researchers also use neuroimaging methods, such as fMRI and event-related potentials (ERPs). These techniques allow

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for a more objective measurement of brain processes related to attention hold and have been found to be more reliable than behavioral measures alone (White et al., 2016).

Neural Basis of Attention Hold

There have also been some studies that have tried to find the neural basis of holding of attention. According to those studies, the left lateral prefrontal cortex (LPFC), as the center of the stream for holding of attention, plays a role in inhibiting attention from disengaging (Ozaki & Ogawa, 2009). Also, during attention-holding tasks, a causal flow between the left thalamus and the left middle frontal gyrus (MFG) has been observed in fMRI (Akimoto et al., 2013). This causal flow was determined by using Granger causality analysis, a statistical method used to infer causal relationships between time series data. In another study, Kelley et al. (2008) found that the sustained activity in lateral regions of posterior parietal cortex and precentral sulcus represents sustained states of peripheral attention. The voluntary hold of spatial attention is also associated with regions located more towards the sides of the parietal cortex, such as the intraparietal sulcus (IPS) and also, the temporoparietal junction (TPJ) (Corbetta et al., 2000; Kim & Hopfinger, 2010; Meyer, et al, 2018; Vandenberg et al., 2001; Yantis et al. 2002). Also, a study has shown that no attentional hold effects were detected during the initial stages of visual processing (<400 milliseconds after the stimulus appeared), which suggests that the hold of attention is linked to alterations in connections between brain regions, rather than changes in the activation levels within specific regions (Schupp et al., 2007). In other words, the absence of attentional hold effects during the initial stages of processing, especially when basic visual processing is underway, indicates that the manner in which attention is held might be more reliant on how various brain regions interact with one another, rather than solely depending on the activity levels of individual regions. This

insight adds complexity to our understanding of how attention works and emphasizes the importance of the connections between different brain regions in holding attention over time.

Given that hold of attention is understudied compared to other aspects of attention (e.g. capture of attention), it remains an important aspect to investigate. Each of these attention functions (capture and hold) happens at a different and specific time in the brain and many studies have tried to investigate these timings for different stimuli (Bardeen et al., 2017; Shang et al., 2021; Song et al., 2022; Torrence et al., 2017). For example, in their investigation of threat-related attention bias variability, Bardeen et al. (2017) observed a rapid covert shift of spatial attention towards angry expressions, acknowledged as threatening stimuli. Using a modified dot-probe task with both neutral and threatening stimuli and employing four distinct stimulus presentation durations, the researchers explored attentional biases related to emotion dysregulation. They calculated Attention Bias Variability (ABV) as a metric assessing within-person variability of attentional bias towards and away from threat stimuli. This measure, capturing dynamic shifts in attention between threat and non-threat stimuli over time, offers insights into how individuals shift their focus, alternating between potential sources of threat and other stimuli. Further investigations by Torrence et al. (2017) explored attention hold and capture for fearful and happy faces using different timings (133, 266, 532 milliseconds) by using the dot-probe paradigm. In their study, fearful faces captured attention within a range of 84 to 266 milliseconds, while happy faces exhibited this effect at times spanning from 168 to 336 milliseconds. Shang et al. (2021) extended these findings, demonstrating attentional bias to fearful faces and revealing attentional effects induced by fearful emotions at a short SOA (stimulus onset asynchrony) of 300 milliseconds, consistently leading to improved visual acuity. In a subsequent 2022 study by Song et al., the focus shifted to positive attention bias in socially anxious individuals. Notably, negative stimuli (angry

and fearful faces) exhibited heightened attention at 100 millisecond SOA, whereas positive stimuli (happy faces) showed increased attention at 500 millisecond SOA. This distinction implies varying involvement of early and later attentional processes in response to positive and negative emotional stimuli.

Event-Related Potentials measures of attention

One way to examine the neural mechanisms of cognitive functions with high temporal resolution is by using event-related potentials (ERP), drawn from electroencephalography (EEG) (Böckler & Sebanz, 2012). ERPs are scalp-recorded voltage changes that show moment to moment alterations in brain processing and, as a result, offer a way to assess the temporal history of activity associated with various phases of perception and can be used to look at how both task-relevant and task-irrelevant stimuli are processed (Hickey et al., 2006). ERPs are connected to particular internal or external events (e.g., stimuli, responses, decisions), providing data regarding a wide range of cognitive and affective processes, which can be recorded noninvasively from practically any set of research participants (Luck & Kappenman, 2012). Benefiting from its high temporal resolution, using ERP allows us to trace the neural basis and the time-course of cognitive processes that occur rapidly (Wójcik et al., 2018).

There are a number of ERP elements that have been connected to attentional processes. For example, P1 (positive deflection at 100 milliseconds after stimulus onset), which is typically the first major visual ERP component, responds to the degree of attentional focus on a stimulus location or feature (Böckler & Sebanz, 2012; Luck, 2014; Shin et al., 2010). Some researchers who have tried localizing the P1 component by using mathematical modeling and fMRI suggest that the initial part of the P1 wave originates from the dorsal extrastriate cortex, located in the

middle occipital gyrus, while a subsequent portion originates more ventrally from the fusiform gyrus (Di Russo et al, 2002; Luck, 2014). The P1 wave is responsive to changes in stimulus parameters, which aligns with its probable origins in the extrastriate visual cortex (Foxe & Simpson, 2002). It is also influenced by selective attention and the individual's level of arousal (Hillyard, Vogel, & Luck, 1998; Luck, Woodman, & Vogel, 2000; Vogel & Luck, 2000).

The P1 wave is followed by the subsequent N1 wave (Luck, 2014). The N1 ERP component, an early component peaking around 100-150 ms after stimulus onset, is closely linked to stimulus filtering and the involuntary orienting of attention (Ghani et al., 2020; Luck, 2014; Näätänen and Michie, 1979; Takeda et al., 2016). It is also thought to signify stimulus detection and possibly play a role in later stages of sensory processing, working in conjunction with subsequent ERP components (Fogarty, Barry, & Steiner, 2020).

Another attention-related component is P3, a positive deflection in the ERP waveform that happens about 300 milliseconds after the start of a stimulus, which is involved in the distribution of attention (Böckler & Sebanz, 2012; Luck, 2014). The P3 component, also referred to as the late positive component (LPC), is primarily localized along the midline of the scalp, with its amplitude increasing from frontal to parietal regions (Johnson, 1993; Luck, 2014; Polich, 2012). The P3 is typically elicited in an oddball paradigm (Donchin et al., 1978). In this paradigm, two different stimuli (e.g., tones) are presented to the participant, with one being more frequent (standard) and the other less frequent (deviant). The P3 is robust when the participant distinguishes the infrequent target from the standard stimulus (Polich, 2012). Additionally, the P3 is sensitive to the level of attentional resources involved during dual-task performance (Luck, 2014). In a dual-task design, participants perform a primary task that varies in cognitive demands while simultaneously engaging in a secondary task that involves mentally counting target oddball stimuli. When the

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primary task is easy, the P3 amplitude associated with the secondary task is relatively large. However, during more difficult primary tasks, the P3 amplitude decreases because more cognitive resources are allocated to the primary task (Wickens et al., 1985).

One ERP component that may be especially related to attention hold and is known for the disengagement and reorienting of attention is Ipsilateral Invalid Negativity (IIN), which is a negative-polarity ERP component observed over temporal-occipital regions of the hemisphere ipsilateral to an invalidly cued target, typically appearing around 200-250 ms after the target onset (Hopfinger & Mangun, 2001). This component is evident only when targets appear at un-cued locations, reflecting attentional reorienting through the disengagement mechanism that shifts attention from the previously cued location to refocus on a new, unexpected target (Posner et al., 1984). This effect is most prominent at shorter cue-to-target intervals, as longer intervals may allow attention to decay or inhibit at the cued location, reducing the need for disengagement (Ford & Hopfinger, 2013). Neuroimaging and patient studies have linked the IIN to activity in the posterior superior temporal gyrus, specifically the temporal-parietal junction, a region implicated in attentional reorienting (Corbetta et al., 2000; Chambers et al., 2004). Research suggests that the IIN may function independently of earlier visual processing components, as it reflects later attentional processes involved in shifting focus rather than initial perceptual biases (Hopfinger & Ries, 2005).

Sustained Posterior Contralateral Negativity

A critical ERP component associated with later stages of information processing and central to understanding attention dynamics is sustained posterior contralateral negativity (SPCN), also known as contralateral delay activity (CDA), contralateral negative slow wave (CNSW), and

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contralateral search activity (CSA) (Chen et al., 2022; Rashal et al., 2023). SPCN is the ERP signature of active maintenance of a laterally displayed stimulus in visual working memory, beginning about 300–650 milliseconds after stimulus onset (Chen et al., 2022; Rashal et al., 2023; Wójcik et al., 2018). Although variously named, SPCN, CDA, CNSW, and CSA fundamentally reflect the same neural processes associated with the maintenance of spatial target representations in visual short-term memory. The similar scalp topographies and time courses of these posterior asymmetries support the idea that they are all part of the same continuous processing system, which re-processes visual information until a stable target percept is created in visual short-term memory to aid in goal-directed behavior (Sander, Werkle-Bergner, & Lindenberger, 2011; Schneider, Hoffmann, & Wascher, 2014). SPCN was initially investigated through cued change detection tasks, where participants were prompted to memorize objects presented in a visual hemifield for later comparison (Chen et al., 2022). This process of retaining information in visual short-term memory is closely linked to holding selected stimuli in the spotlight of attention, known as sustained visuospatial attention (Holmes et al., 2014).

Research has shown that SPCN's amplitude is sensitive to the number of items maintained in visual working memory and correlates with individual working memory capacity (Luria et al., 2016; Adam, Robison, & Vogel, 2018). Therefore, studies show that this ERP marker is an indicator of visual working memory load and provides insight into both stable visual representations and fluctuating attentional resources during memory retention, making it a valuable tool for understanding how attentional mechanisms support visual cognition (Chen et al., 2022; Ikkai, McCollough, & Vogel, 2010; Lefebvre et al., 2011; Roy, & Faubert, 2023).

In a study investigating the attention capture, hold, and bias where the participants were presented with the pictures of their own face during a dot-probe task, Wójcik et al. (2018)

hypothesized that the presence of *attentional hold* effects can be measured by SPCN. In their study with self-face cues, Wójcik et al. (2018) found that SPCN was exclusively present for longer (600 milliseconds) self-face cue presentations. They explained that the involuntary self-face processing prevents the top-down control setting from filtering out distractors, leading to the engagement of visual short-term memory. Therefore, it was concluded that the presence of SPCN indicates an attentional hold effect. In their opinion, keeping the visual short-term memory information about the salient stimuli is thought to drive top-down template signals that facilitate the hold of visual spatial attention, even when they are not task-relevant (Wójcik et al., 2018). Moreover, both MEG (magnetoencephalography) and fMRI recordings agree that the neural generators of SPCN are situated in the parietal cortex, particularly in the intraparietal sulcus (Chen et al, 2022).

SPCN is often observed following the N2pc, which is another ERP component that is used for studying attention capture (Holmes et al., 2009; Rashal et al., 2023). The N2pc is a negative deflection in the ERP waveform, which is detected approximately 200–300 milliseconds after stimulus onset at parieto-occipital sites, contralateral to the visual hemifield in which the target is displayed (Chen et al. 2022; Luck, 2014; Wójcik et al., 2018;). It is used to determine the shift of focus of attention to the location of a salient stimulus within a visual scene (Luck, 2014; Wójcik et al., 2018). N2pc is named for its polarity, latency, and topography, which is the posterior contralateral negativity in the latency of the N2 component and is strongly related to attentional processes (Hickey et al, 2006). The N2pc signifies the process of suppressing distractors, particularly when visual distractors complicate the processing of a visual target. However, it is conceivable to observe substantial N2pcs even when the target for processing is the sole visual stimulus in a given visual field, accompanied by a solitary distractor in the opposite visual field (sometimes with a fixation point). Researchers generally agree that the N2pc reflects distinct

processing of stimuli in one visual field compared to those in the opposite visual field, and this processing signifies the operation of visual-spatial attention. The term "posterior contralateral negativity" (PCN) is occasionally used to describe the N2pc, emphasizing its functional distinction from the typical N2 component (Jolicœur et al., 2008). Additionally, more lateral/ventral regions, which are also implicated in the generation of N2pc activity, play a role in the generation of SPCN activity (Chen et al., 2022).

Despite the apparent similarity between SPCN and N2pc, SPCN emerges later (around 400 ms) and persists significantly longer than N2pc. Specifically, it lasts as long as the retained objects are held in visual working memory (Chen et al., 2022; Luria et al. 2016; Vogel, & Machizawa, 2004). The amplitude of SPCN increases with the number of objects to be retained in memory, as long as this number does not exceed an individual's visual working memory capacity, which typically averages around 3 objects across individuals (Chen et al., 2022). In another study, when the subject's objective was to process a red item at fixation, a red item in the periphery captured visual-spatial attention, eliciting an N2pc (Jolicœur et al., 2008). However, subjects succeeded in excluding this item from visual short-term memory, resulting in the absence of an SPCN following the N2pc. Despite using stimulus displays similar to those where both N2pc and SPCN are observed, the absence of SPCN occurred when the task involved encoding the peripheral target. These findings offer additional supporting evidence for a functional dissociation between the N2pc and the SPCN. Although a top-down control mechanism effectively screens out distractions to block entry into visual short-term memory, it inadequately prevents the seizing of visual-spatial attention by stimuli that share the target-defining feature, resulting in an N2pc. This suggests that the SPCN is closely linked to stimulus processing rather than response processing. In essence, while the N2pc reflects the allocation of attention towards the location of a target stimulus, the

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SPCN seems to be more directly tied to the cognitive processing of the stimulus content itself, particularly its encoding and maintenance in visual working memory (Jolicœur et al., 2008). Figure 1 shows the two components' different timing and waveforms in Wójcik et al.'s study (2018).

Maintaining attention on cues related to threats may also tie into the retention of such information in visual short-term memory, contributing to the SPCN. Consequently, the current findings regarding the SPCN may indicate the activation of neural mechanisms associated with holding attention on both internal and external representations of threat (Holmes et al., 2014). The SPCN as an extension of the N2pc, reflects the continual allocation of attention resources to the visually biased hemifield. This sustained spatial attention may be influenced by top-down template signals from visual short-term memory (Holmes et al., 2009).

In contrast to the N2pc, which maintains a consistent amplitude regardless of memory load, the amplitude of the SPCN varies in accordance with visual short-term memory load and persists throughout the entire retention interval. Notably, studies have demonstrated that the SPCN amplitude is responsive to changes in stimulus set size within a brief timeframe, potentially explaining why investigations with extended exposure times might not capture the period of heightened SPCN amplitude (Tseng et al., 2012). Despite the variability of SPCN during the retention interval based on memory load, effective filtering of irrelevant information can prevent the SPCN from prematurely reaching its asymptotic level by reducing the usage of attentional and memory resources for unnecessary processing. Consequently, the crucial factor distinguishing high and low performers may lie in their ability to prioritize relevant information and suppress distractors (Tseng et al., 2012). Notably, a wave form similar to SPCN was detected even in trials where the target was absent, suggesting potential working memory engagement linked to the search for a target and/or processes associated with confirming its absence (Kiss et al., 2008).

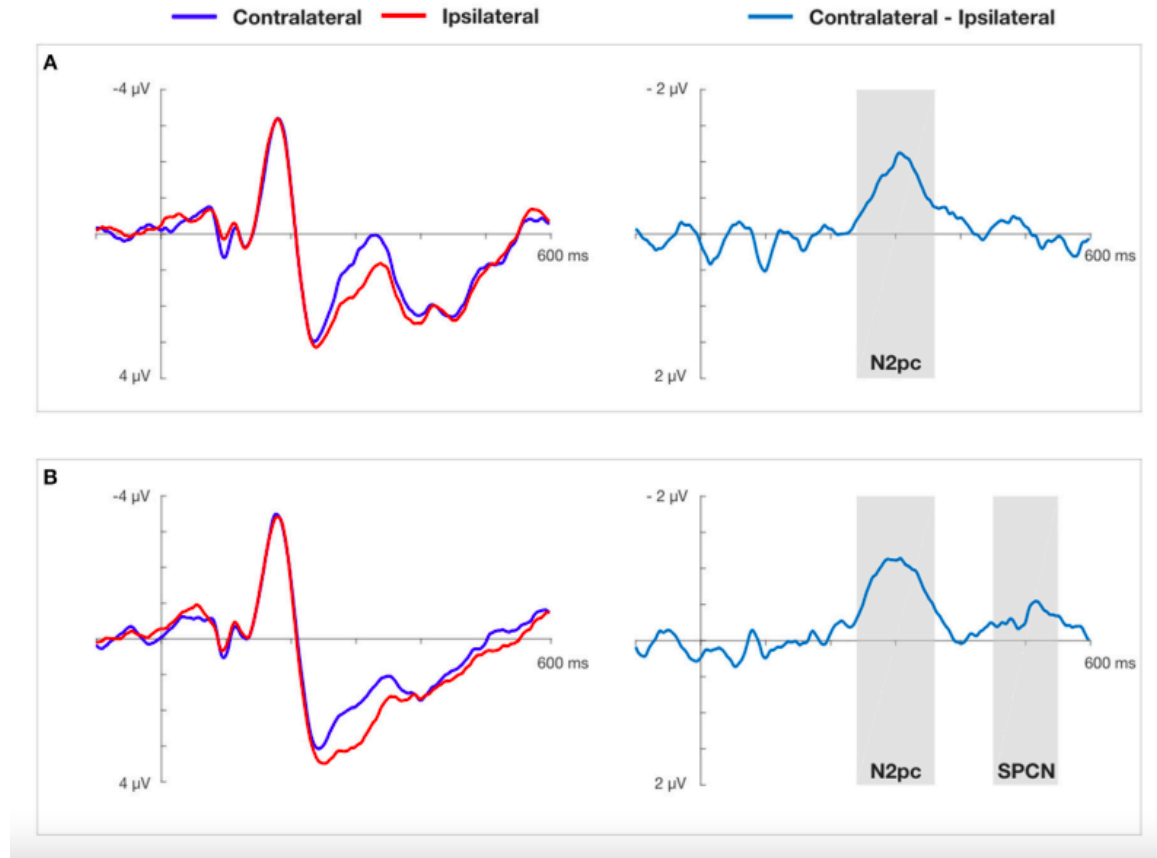


Figure 1. Grand average ERPs time-locked to the onset of the cue-faces (self vs. other) at electrodes PO7/PO8 and difference waves for 150 ms (A) and 600 ms (B) cue presentation time. From *Self-face captures, holds, and biases attention*, by Wójcik et al, 2018.

It is proposed that the SPCN indicates ongoing processing of the attended item. For instance, an increase in SPCN amplitude is observed when the difficulty of the discrimination task is heightened (Mazza et al., 2007; Prime & Jolicœur, 2009). As mentioned before, it also serves as a valuable indicator of cognitive functions associated with holding information in visual short-term memory (Chen, 2022). These functions are presumed to generate top-down template signals that facilitate the maintenance of visual spatial attention directed towards stimuli relevant to a specific goal (Holmes et al., 2009). Studies have revealed that reproducing the SPCN waveform

involves holding more complex stimuli in visual working memory, resulting in heightened amplitudes, owing to the informational load on working memory (Wójcik et al., 2018).

Face Stimuli

One of the salient stimulus that has been used in many studies is human faces, which have the potential to capture and hold our attention (e.g., Chanon & Hopfinger, 2011; Chen et al., 2021; Dou et al., 2021; Eimer & Holmes, 2007; Fajkowska et al., 2011; Lassalle & Itier, 2013; Maxwell et al., 2022; Wójcik et al., 2018; Yrttiaho et al., 2022). Faces play an important role in social interaction, expressing essential information about identity, emotions, and intent (Nakajima et al., 2012). Therefore, faces could be a great tool for examining how attention is allocated. Faces are considered to receive benefit from general attentional prioritization (Doradzińska et al., 2022). As an example, adults are faster at identifying and classifying faces than any other objects (Parks et al., 2014). Also, face-specific processing is present early in infancy (Luck, 2014), in such a way that infants prefer to look at human faces, suggesting that face recognition might be innate (Nakajima et al., 2012). Researchers also have shown that human faces are processed more effectively than animal faces, implying that there are no broad detection windows for all face-like patterns, but that human adults process their own species' faces more effectively than any others (Simpson et al., 2014). Also, people will still read information from the face, even though surrounding information (e.g. body gestures) are present (Jenkins, 2017).

Some studies have also shown brain activity directly resulting from face processing. Previous work using functional magnetic resonance imaging (fMRI) has shown that the fusiform face area (FFA) is primarily activated by visual stimuli of faces (Haeger et al., 2021). Also, in ERP studies, the N170 component – a negative ERP component recorded between 140 and 200

milliseconds from stimulus onset, peaking around 170 milliseconds at the posterior temporal regions – clearly distinguishes facial from non-facial visual stimuli, resulting in considering N170 as a face-selective component (Nakajima et al., 2012). In addition, research on neurodevelopmental diseases has revealed that children with autism spectrum disorder have abnormalities in the ERP waveform's N170 component, which supports the role of N170 in face detection (Luck, 2014). Finally, N170 is insensitive to face familiarity, gender, and race suggesting that it is not involved in face identification (Nakajima et al., 2012).

In the studies on capture and hold of attention between faces versus places (e.g., Black, 2015; Parks et al., 2014), researchers found two independent mechanisms that contribute to distraction: an initial involuntary capture to any unexpected onsets, followed by a subsequent holding of attention to a potentially significant, but task-irrelevant stimulus, which in this case was a human face. Interestingly, in their study, Parks et al. (2014) found that the later holding of attention by faces did not occur just for fearful faces, but also for neutral ones.

Motivation for the Current Study

Only a few studies have interpreted SPCN as a measure of attention-holding (Holmes et al., 2009; Wójcik et al., 2018). These studies suggest that SPCN may reflect sustained engagement with stimuli rather than the shifts in attention typically associated with attentional capture. However, they did not include robust behavioral measures to confirm that attention was truly held, rather than merely captured, when an SPCN waveform was observed in their dot-probe paradigms. This gap leaves the link between SPCN and attentional hold tentative and underscores the need for further investigation.

The present study builds on these findings by proposing that SPCN, traditionally linked to working memory, can also serve as a marker of attentional hold. Working memory comprises domain-general executive attention coupled with domain-specific short-term memories (Ngiam, 2024). This dual mechanism suggests that SPCN, a neural marker of working memory, could reflect attentional hold by signaling the sustained engagement of attention with relevant stimuli. Recent findings further support this perspective by showing that poor attentional ability, often associated with low working memory capacity, stems from deficits in overriding involuntary attentional capture signals from distracting information (Chota et al., 2022). This ability to override attentional capture and sustain attention on target locations highlights a critical interaction between voluntary and involuntary attentional processes. Specifically, the mechanisms underlying attentional hold appear to blend goal-directed, voluntary control with the automatic prioritization of stimuli that partially match control settings (Chota et al., 2022; Rutkowska, Doradzińska, & Bola, 2022).

By using a dot-probe task shown to evoke SPCN responses (Holmes et al., 2009; Wójcik et al., 2018) and simultaneously recording EEGs, this study seeks to test whether SPCN can reliably serve as an ERP component for measuring attentional hold. Establishing this association would not only fill an important gap in cognitive neuroscience but also advance our understanding of how attentional hold operates through sustained neural mechanisms. This study's findings may clarify whether SPCN can reliably index attentional hold and provide new insights into the interaction between voluntary and involuntary attentional processes (Chen et al., 2022; Rashal et al., 2023).

The information we learn from this study will help us unravel different aspects of attentional hold and what factors could help us maintain our attention to a stimulus or even stop attending to

it. The functions of attention are critical for many everyday tasks, from avoiding distractions while driving to focusing on school or job-related work. The present study aims to provide insights into understanding the hold of attention by specifying the brain mechanisms that underlie this function. This research is important because its findings can enhance our understanding of how attention operates. By identifying the neural markers of attention-holding in ERP components, we may grasp how to improve attention hold.

We plan to examine attention hold by using faces and places, based on the result of my first-year project. As mentioned earlier, the previous work has shown that faces have the potential to capture and hold our attention (Maxwell et al., 2022; Yrttiaho et al., 2022). For this study, the reason behind choosing images of places along side with faces, is that they constitute a distinct class of extremely complex objects (same as facial cues) whose perception has been linked to specialized brain circuits (Parks et al., 2014). By images of a place, we mean a view of the environment that comprises both background elements and discrete objects, all arranged according to certain spatial and semantic rules (Doradzińska et al., 2022).

In our study, we wanted to make sure that attention is actually held and not just captured. Torrence et al. (2017), using the dot-probe task, had shown that fearful facial expressions rapidly and fleetingly capture attention within a brief timeframe. Happy faces, on the other hand, follow a longer time-course, and they also capture and hold our attention. But in comparison with the negative emotions, the attention does not rapidly disengage from them (Torrence et al., 2017). Therefore, we used happy faces in our study to measure attention hold in the visual dot-probe task. However, in their study, Torrence et al. used only behavioral measures of attention capture and hold, without any neuroimaging measures to identify neural markers or distinguish between these two attentional functions. As mentioned earlier, behavioral measures provide valuable insights but

still have limitations, making it preferable to combine them with neuroimaging measures for a more comprehensive approach.

In our recent behavioral experiments, we investigated the timings of the dot-probe task for attention hold in order to find out the exact timings that the attention is not only captured but also held to the presented stimuli. Also, another goal of these experiments was to investigate how the type of noise used in the dot probe tasks and how it related to the target could have an effect on the capture and hold of attention. In those experiments, using a dot-probe paradigm, the participants were shown four different image pairs: a face and a place, a face and a noisy scrambled square, a place and a noisy scrambled square and two noisy scrambled squares, which worked as the baseline (see Figure 2 for the different conditions). The images of faces were all happy female faces with a forward gaze, drawn from the Chicago Face Database and gray scaled. We also used the places images (which were exterior views of houses) because they had the same amount of complexity to facial cues and were used in the previous studies as well (Parks et al., 2014; Song & Kim, 2024; Yi et al., 2004). The scrambled noisy squares were the scrambled pixels of the two images of faces and places randomly combined and used as the baseline. Each of these conditions were shown for 3 different timings. In our first experiment, the images were shown for either 100, 300, or 500 ms. Having four different conditions with three different timings made the task having 12 unique conditions. Each of these conditions were shown 80 times throughout the whole experiment. The experiment included 8 blocks, each having 120 trials, which means each participant had done 960 trials in total. Completing the whole task took 50 minutes approximately, with a rest between each block, and the participants were given one hour credit after participation through SONA.

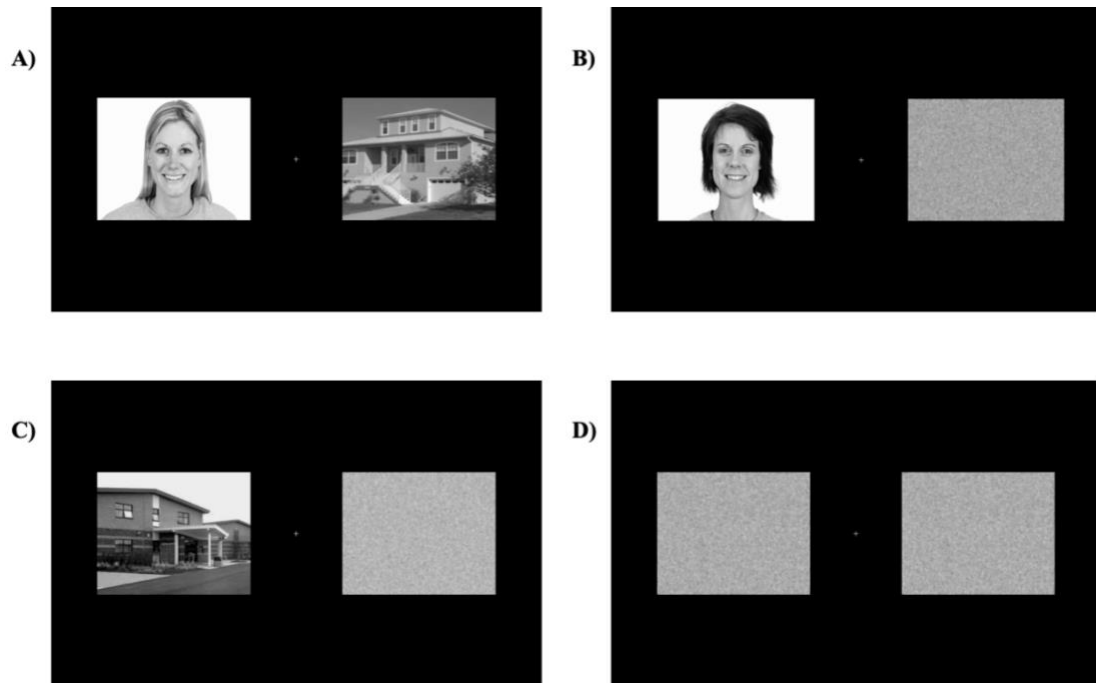


Figure 2. Examples of the four conditions in the dot-probe task. The conditions used in the experiment are: A) face-noise and B) place-noise,. The conditions C) face-place and D) noise-noise. Note: Conditions A and B are the primary focus of this study, as they allow us to investigate whether faces and places differentially capture and hold attention. Conditions C and D are included for comparison and the base line, but are not the main focus of the analysis. In the actual experiment, the location of the images and noise stimuli (left vs. right side of the screen) was counterbalanced across trials to avoid positional bias and ensure equal likelihood of presentation on either side.

Each trial started with a central white fixation cross (which was always present throughout the experiment) on a black background. After 500 ms one of the mentioned pairs of images was displayed. The time of the display was either 100, 300, or 500 ms in our first experiment. After the offset of the images, a probe (which was an asterisk) was presented until a response was made or until 1.5 seconds had passed. There was a 54 ms gap between the image pair offset and onset of target. There was an inter trial interval of 750-1250 ms randomly assigned between each trial. This trial sequence was used as previous studies have shown that it is particularly sensitive to attentional bias (Holmes et al., 2009). Participants were instructed to press one of two buttons on the keyboard,

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using the index finger and middle finger of their right hand, to indicate as quickly and as accurately as possible the location of target (i.e., left or right). Figure 3 shows an example of the trial sequence.

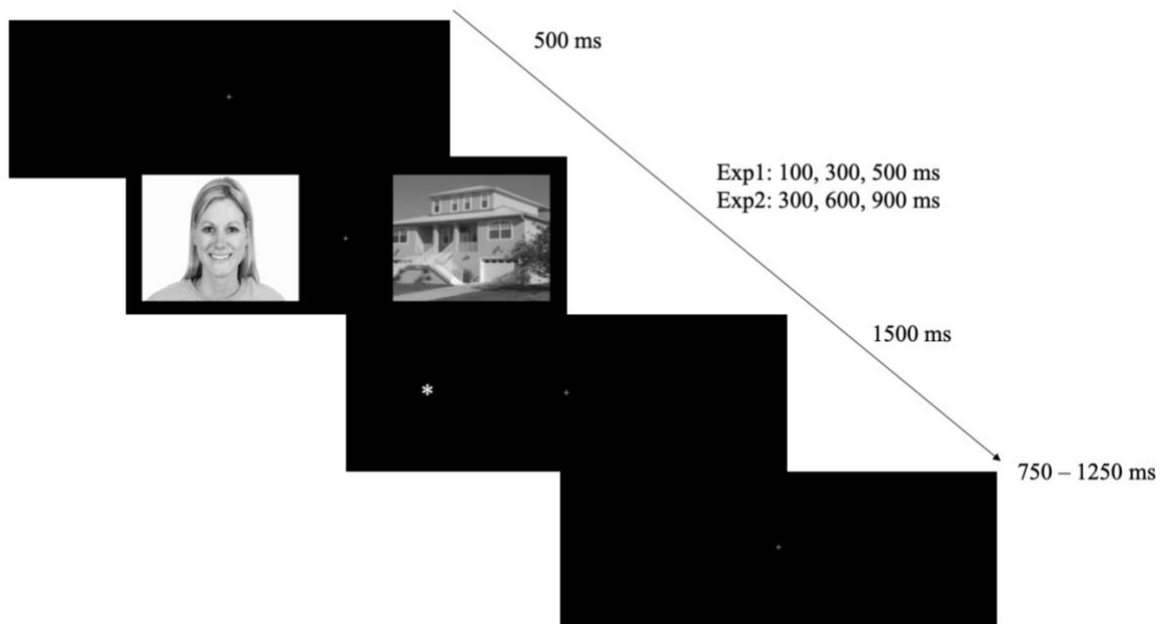


Figure 3. Example of one trial sequence of the two experiments.

Participants were also asked to keep their gaze focused on the central fixation location throughout the experiment. The task had 32 practice trials, where participants received feedback on both reaction time and accuracy for each trial. But in the subsequent experimental blocks, feedback was not provided. All trial types were equiprobable across each of eight experimental blocks. The different trial types were presented with the position of the cues and the position of the probe counterbalanced across trials, so that each appeared in either location with equal probability. All different trial types were presented in a fixed but random order for each participant within each block.

For our second experiment, all conditions were kept the same as in the first one, with the exception of using longer SOAs of 300, 600, and 900 ms. The decision to conduct this second experiment was driven by the need to explore attention-holding processes more effectively. In the first experiment, the shorter SOAs did not reveal much differentiation in attention hold between the conditions, limiting our ability to draw conclusions about sustained attentional engagement. By increasing the SOAs in the second experiment, we aimed to better capture the dynamics of attention hold over longer timeframes.

According to the results of those behavioral experiments, attention was captured to the face stimuli when it was presented for 100 milliseconds, and it was held for 300 and 500 ms as well (Figure 3). This was applicable for the conditions with one face and one noise. However, in our other study where we changed the SOAs to 300, 600, and 900 ms, we did not see such an effect for the face stimuli. But the place-noise conditions were the ones that were capturing attention at 300 ms and this attention orienting was held for the longer SOAs as well (Figure 4). It is noteworthy that for the experiment with shorter SOAs, attention was also captured for places at 300 and was held for 600 ms as well, but it was not captured at the 100 ms. This finding is in line with a recent study that found that typically developing children show attentional capture and hold for place stimuli rather than face stimuli when the context of distractors was inconsistent, having both a face and a place in the conditions (Song & Kim, 2024). As mentioned before, according to the literature SPCN is a late ERP component that happens 300-650 ms after the onset of a stimuli. Also, SPCN was evoked for the 600 ms conditions in another dot-probe study and not the 100 ms SOAs (Wójcik, et al., 2018). Therefore, we will use the longest SOA (i.e. 900 ms) to examine this ERP component and the attention holding in the dot-probe task.

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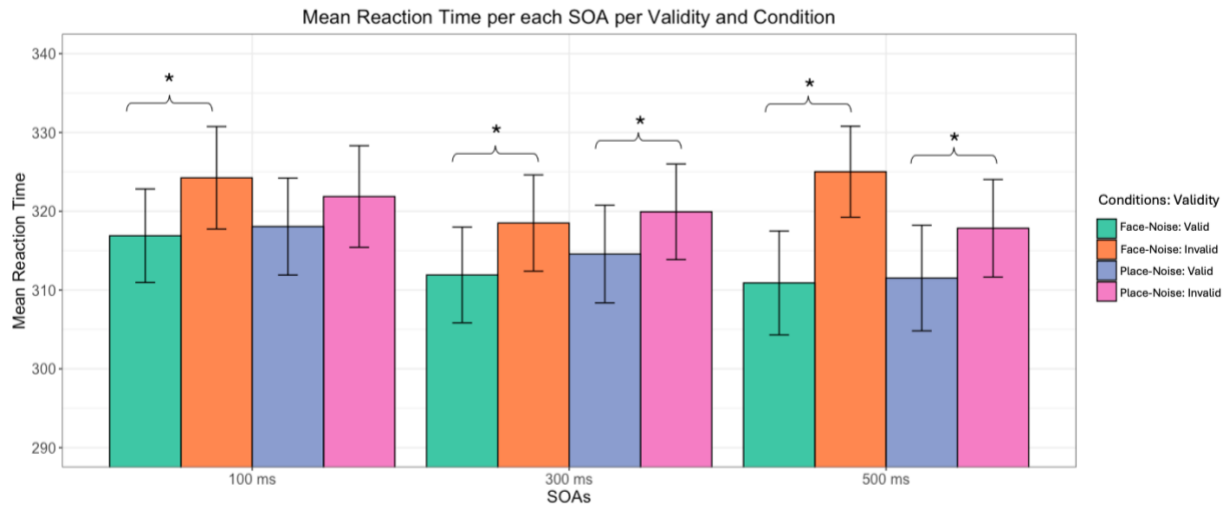


Figure 4. Mean Reaction Times for each SOA (100, 300, and 500 ms) in Face-Noise and Place-Noise conditions per the validity. Note that valid in this plot refers to the times that the probe was shown after a face image or a place image in the face-noise and place-noise conditions, respectively; while invalid refers to the times that the target was shown after the noise in the conditions.

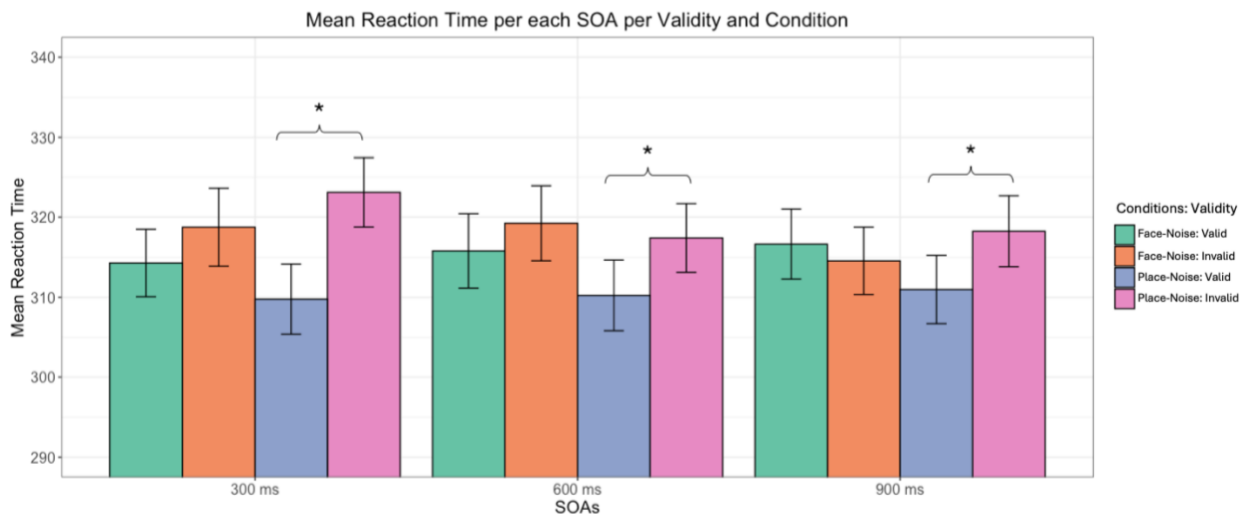


Figure 5. Mean Reaction Times for each SOA (300, 600, and 900 ms) in Face-Noise and Place-Noise conditions per the validity. Note that valid in this plot refers to the times that the probe was shown after a face image or a place image in the face-noise and place-noise conditions, respectively; while invalid refers to the times that the target was shown after the noise in the conditions.

As mentioned above, in our two experiments, we observed that in face-noise conditions, initial attentional capture occurred at 100 ms; however, attention was not maintained on faces for longer SOAs. This pattern is consistent with some findings in the literature (e.g., Puls & Rothermund, 2018; Song & Kim, 2024) but contrasts with others (e.g., Parks et al., 2014; Torrence & Troup, 2017). Therefore, these observations suggest a possible suppression mechanism in the brain that prevents sustained attention on face distractors over time.

The signal-suppression model suggests that while physically salient features of a stimulus initially attract attention, inhibitory processes can override this attentional capture by suppressing the salient stimulus (Gaspelin et al., 2015; Gaspelin & Luck, 2018a, 2018b, 2018c; Sawaki & Luck, 2010). Early EEG research provided evidence supporting the role of suppression in attentional allocation. When presented with a salient but irrelevant distractor, researchers observed an increase in positive amplitude, identified as the distractor positivity (P_D) component, which indicates suppression (Hickey et al., 2009; Sawaki et al., 2011; Sawaki & Luck, 2010). The identification of this component contributed to the hypothesis that attentional suppression plays a role in managing distractors. P_D is typically measured between 250 and 300 ms after stimulus onset, relative to a 100-ms pre-stimulus baseline (Drisdelle & Eimer, 2021; Sawaki & Luck, 2011). However, recent ERP research has proposed an alternative interpretation; Kerzel and Huynh Cong (2023) found that the P_D response to distractors might instead reflect an N2pc component, directed toward an opposing stimulus. This suggests that rather than actively suppressing distractors, the attentional system may instead redirect focus to an opposite location, thereby avoiding the distractor.

Based on our behavioral results and the absence of prolonged holding effects in face-noise conditions at longer SOAs, a secondary objective of our study is to investigate whether EEG data

will show an N2pc response, indicating attentional redirection, or a P_D component, reflecting suppression, in the face-noise conditions. We hypothesize that, over longer presentations, participants engage in suppression of these stimuli. Although we might be seeing an N170 component (indicator of face stimuli) and N2pc (indicator of attention capture) in the first few trials of the experiment, as we approach through the task, the participants might be able to suppress the salient distractor (i.e. the face images) in the following trials and showing a P_D component as they continue responding to the conditions. We might also observe that N2pc present for the noise images in the face-noise conditions as Kerzel and Huynh Cong (2023) explained the suppression mechanism.

Additionally, we might observe an IIN component in the dot-probe task. As mentioned earlier, the IIN component is relevant for examining attentional reorienting, as it reflects the brain's response to unexpected targets appearing at un-cued locations. Typically appearing 200-250 ms after target onset, the IIN indicates the process of disengaging attention from a previously cued spot to refocus on a new target (Hopfinger & Mangun, 2001; Posner et al., 1984). In my study, we would expect the IIN component to appear primarily in conditions where targets appear in un-cued or unexpected locations. In our task, this corresponds to the probe appearing after a noise image (invalid conditions). This expectation aligns with findings from studies involving attention to motivationally significant cues, where the IIN reflects an attentional shift from a cued location to a new, unexpected target (Hopfinger & Mangun, 2001; Shin et al., 2010), indicating that participants needed to disengage from their initial focus on cued locations. Similarly, in our face-noise or place-noise conditions, if participants are initially drawn to the cued location, the IIN component would reflect the need to disengage and reorient attention to the opposite side when

the target appears unexpectedly. For valid conditions, where targets appear at the cued location, the IIN would likely be absent, as no reorienting would be necessary.

To summarize our research questions, we aim to explore the following hypotheses:

1. Our primary goal is to investigate whether SPCN serves as a reliable neural marker of attention hold. To test this, we hypothesize that at the 900 ms SOA, the SPCN component will be observed in conditions that show behavioral evidence of hold (i.e., the place-noise conditions) compared to conditions without behavioral evidence of hold at the 900 ms SOA (i.e., the face-noise conditions).

2. If P_D is the indicator of suppressing a salient distractor, the face images will show that component in the face-noise condition at 900 ms SOA compared to the same condition at 100 ms SOA.

3. If N2pc is the indicator of attention capture, we hypothesize that we will see this component for the face-noise and place-noise conditions in both the 100 ms and 900 ms SOAs, contralateral to the faces or places images. We expect attention to be captured by both the face and place stimuli in both SOAs, but the impact on reaction times may differ, with the 100 ms face-noise condition showing a stronger effect. However, for the 100 ms place-noise condition, the effect may be less pronounced, as the target may not be fully processed in such a brief time window.

4. If IIN is the indicator of disengaging attention, we expect to see this component in invalid trials of both the face-noise and place-noise conditions, particularly in the longer SOA trials (900 ms). Invalid trials refers to the ones where the target is shown where the noise image previously was in the face-noise and place-noise conditions. The shorter SOA (100 ms) may not show the IIN due to the overlap with later components and the brief time window for attention disengagement.

Proposed Experiment

Participants

Our sample will include 27 participants that will be drawn from the undergraduate students in the department of Psychology and Neuroscience at University of North Carolina at Chapel Hill, taking part in the PSYC 101 course. The participants will be right-handed, have normal or corrected to normal vision, and have no neurological disorders, and will be given 2-hour credits for their participation. An a priori sample size of 27 subjects was determined based on a power analysis using G*Power version 3.1 (Faul et al., 2009). This analysis focuses on the SPCN, N2pc, P_D, and IIN components, which are essential to testing our hypotheses. Previous ERP studies investigating similar attentional processes using the dot-probe task have used comparable sample sizes, demonstrating reliable effects with samples of around 20-22 participants (Gaspelin & Luck, 2018b; Holmes et al., 2009; Hopfinger, & Mangun, 2001; Wójcik et al., 2018)

Results indicated that a sample of 23 participants would provide 95% power to detect a medium effect at a significance level of $\alpha = 0.05$ with false discovery rate of 0.05 (Bonferroni adjusted), and a repeated measures ANOVA examining the dot-probe task with 8 conditions (2 different image pairs (face-noise and place-noise), 2 different timings (100 ms and 900 ms), and 2 different validity levels (valid and invalid)). Therefore, accounting for a possible rejection of poor ERPs in our data sets (15%), a sample size of 27 is sufficient for a repeated measures ANOVA in this study. Only correct responses that are between 150 and 750 milliseconds after target presentation will be used in the following analyses (Torrence et al., 2017). This study will be approved by the institutional review board at the University of North Carolina Chapel Hill.

Material

Participants will first read and sign the consent form and complete a demographics and handedness questionnaire. In the instructions section at the beginning of the task, participants are explicitly directed to use the index and middle fingers of their right hand for key presses.

EEG Recording and Pre-Processing

EEG will be recorded using the BioSemi ActiveTwo system with active electrodes positioned at the left and right mastoids and 96 electrode sites. Electrooculogram (EOG) recordings will be captured from electrodes placed at the outer canthi of each eye and above and below the right eye to monitor blinks and eye movements. Trials containing eye movements or blinks will be excluded from analyses after offline processing. EEG data will be referenced to the right mastoid, digitized at 256 Hz, and amplified with a bandpass filter of 0.05–100 Hz.

Post-acquisition analyses will be conducted with ERPLab Toolbox (Lopez-Calderon & Luck, 2014) and EEGLab Toolbox (Delorme & Makeig, 2004). EEG signals will be re-referenced to the average of the left and right mastoids, while EOG signals will be processed using bipolar horizontal and vertical derivations. Offline filtering will apply a noncausal Butterworth high-pass filter (0.1 Hz half-amplitude cutoff, slope of 12 dB/octave). Averaged ERP waveforms will be computed over a 1000 ms epoch starting 100 ms before stimulus onset, with temporal precision maximized by a low-pass filter (30 Hz half-amplitude cutoff; 12 dB/octave) to reduce high-frequency noise.

Separate averages will be computed for different combinations of cue types (face-noise vs. place-noise), Validity (valid vs. invalid), and contralaterality (electrode ipsilateral vs. contralateral

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to the cue's location). The ipsilateral waveform will be calculated as the average of electrodes on the same side as the cue, while the contralateral waveform will be derived from electrodes on the opposite side of the cue. The SPCN, N2pc and P_D components will be assessed based on the mean amplitudes of ERPs recorded at PO7 and PO8 within three-time windows (SPCN: 320–500 ms; N2pc: 200–400 ms; P_D: 100–300 ms). Also, The IIN component will be assessed based on the mean amplitude of ERPs recorded at the T5 and T6 electrodes ipsilateral to the target side within the 220–280 ms time window post-stimulus (30 ms before and after the average peak latency). Selection of electrode sites and time windows will be informed by both individual subject waveforms and previous studies (Gasplin, & Luck, 2018b; Holmes et al., 2009; Shin et al., 2010; Wójcik et al., 2018). R Statistical Software will be used to complete statistical analyses (v2021.09.2; R Core Team, 2021). The package rstatix (Kassambara, 2021) will be used for soing the statistical analyses, and the ggpubr (Kassambara, 2020), ggplot2 (Wickham, 2016), ggstatsplot (Patil, 2021) and tidyverse (Wickham, 2019) packages will be used to manipulate data and produce figures. All analyses will be reported at a significance level of $p < 0.05$. Greenhouse–Geisser adjustments to the degrees of freedom will be applied as necessary for all analyses.

Design

The experiment will use a 3 x 2 design in which trial type (face-noise, place-noise, and noise-noise) and image duration (100 and 900 ms) will be manipulated within subjects. The experiment will have eight blocks, with 120 trials per block. There are 3 different pairs of images in this task: a face with a scrambled square, a place with a scrambled square, or two gray squares as the baseline. The images of faces are all happy female faces with a forward gaze, drawn from the Chicago Face Database (CFB) and gray scaled. We are only using faces with forward gaze,

because it has been shown that direction of the eyes of the faces affects the location to which the participant attends (Chanon & Hopfinger, 2011). The images of places are all exterior views of houses (also gray scaled) that had been used by the previous studies (Parks et al., 2014; Song & Kim, 2024; Yi et al., 2004). The scrambled squares are the scrambled black and white images of the face-place pairs and are blurred, which serve as the base line for comparison.

Procedure

We will conduct a behavioral pilot study incorporating modifications compared to previous behavioral studies, using a modified version of the current IRB for these changes. The main differences in this study are that we will only include two SOAs (100 ms and 900 ms), compared to the three SOAs used previously, allowing for better comparisons between conditions for the ERP components of interest. This modification allows us to concentrate on the most critical SOAs, enabling us to maximize the number of trials in these conditions. A higher number of trials enhances the quality and reliability of the ERP components of interest, as ERPs are more robust with larger trial counts. Additionally, we will exclude the face-place conditions used in prior studies, as they do not add value to the comparisons and analyses we plan to conduct. We expect to replicate the results observed in our previous behavioral experiments. While we aim for replication, we recognize that there may be variability in the replication process. If the results deviate from expectations, we will carefully examine the data to identify potential causes and determine what modifications may be necessary. Based on these findings, we will adjust the experimental design and, if needed, run another pilot study to ensure the reliability and validity of our approach. After obtaining these results, we will proceed with the EEG study.

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After getting approval from the UNC Chapel Hill Institutional Review Board (IRB) for the EEG study, the study will be posted on UNC SONA system where the students can have access to the contact information of the researchers to set up a time for visiting the lab and doing the task. After study completion, participants will be compensated for their participation with 2 class credit.

Each trial starts with a central white fixation cross (which is always present throughout the task) on a black background. After 500 ms, one of the aforementioned pairs of images is also displayed. The time of the display of each of those pairs is either 100 or 900 ms. After the offset of the images, a probe (which is an asterisk) is presented until a response is made or 1.5 seconds has elapsed. There is an inter trial interval of 750-1250 ms randomly assigned between each trial. This trial sequence was used as previous behavioral investigations have shown it to be particularly sensitive to attentional bias (Holmes et al., 2009). Figures 6 and 7 depict the experiment's design. Participants are instructed to press one of two buttons on the keyboard, using the index finger and middle finger of their right hand, to indicate as quickly and as accurately as possible the location of target (i.e., left or right). To avoid spatial compatibility conflicts that arise when a subject must make a left-hand response for a right-side target or vice versa (known as the Simon effect), it is usually preferable to have subjects use two different fingers of the same hand to make the two different responses (Luck, 2012).

Participants are also asked to keep their gaze focused on the central fixation location throughout the task. The task comprises 32 practice trials, where participants receive feedback on both reaction time and accuracy for each trial. However, in the subsequent experimental blocks, feedback will not be provided. All trial types are equiprobable across each of eight experimental blocks. The different trial types are presented with the position of the face and the position of the probe counterbalanced across trials, so that each appeared in either location with equal probability.

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All different trial types are presented in a fixed but random order for each participant within each block. This task will take approximately 45 minutes to complete. Participants will also be allowed to rest between each block.

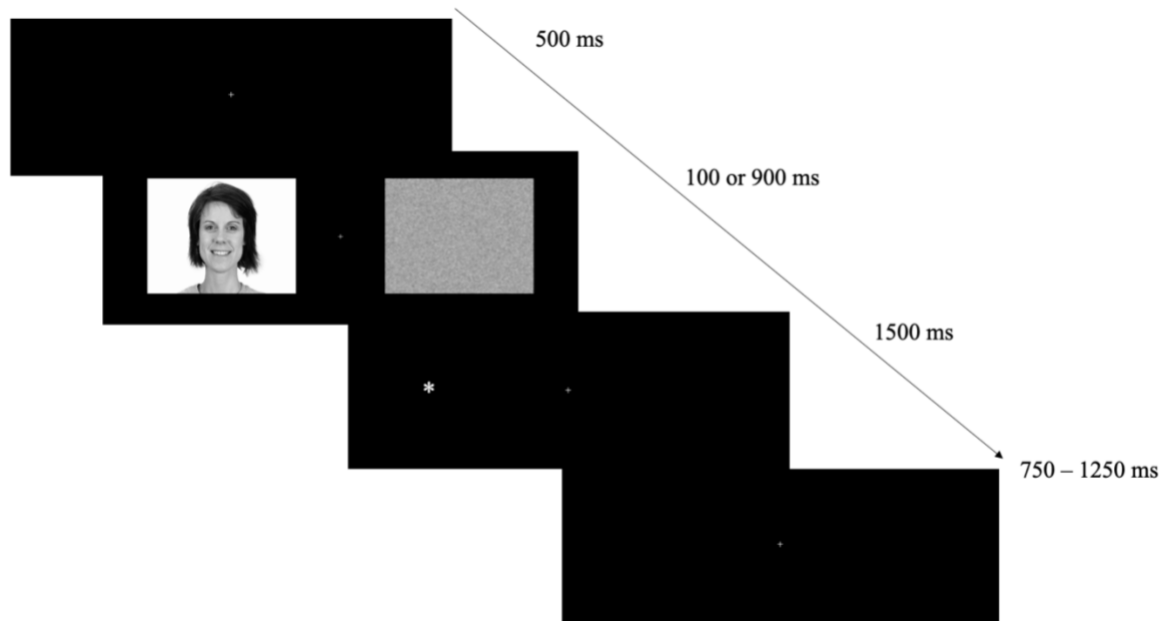


Figure 6. The outline of the task's stages. The Face-Noise condition is shown in this figure as an example

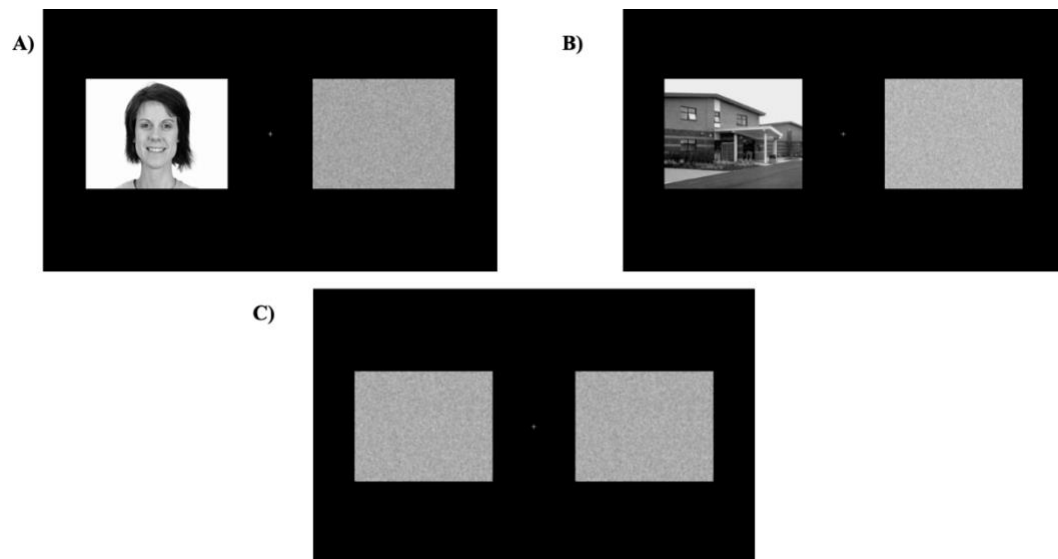


Figure 7. The three conditions of the experiment. A) face-noise pair; B) place-noise pair; and D) noise-noise pair (the base-line condition).

Predicted Results

Behavioral Results

The current design measures continuous variables (reaction time) of two different timings as dependent measures. Reaction Times (RTs) that are two standard deviations below or above the mean as well as error trials will be excluded from the analysis of reaction times. Additionally, participants with near chance responses will be removed from analyses. Due to ceiling effects, and non-normality of data, ANOVA analysis will not be performed on accuracy measures given the lack of variance of the data.

Results will be reported at a significance level of $p < 0.05$ for all analyses. Where appropriate, degrees of freedom will be Greenhouse–Geisser corrected. For the ANOVA, effect sizes will be measured additionally by calculating the partial eta squared, which measures the degree of association between the independent variable and the dependent variable (Cohen, 1973). Regarding the interpretation of partial eta squared we will follow the approximate guideline proposed by Cohen (partial eta squared = .01 small effect, partial eta squared = .09 medium effect, partial eta squared = .25 large effect). Recent studies of attention using this task have similarly used these analyses in publishing results that are important to attention theories (Carlson & Fang, 2020; Hirai et al., 2022; Kaiser et al., 2020; Shang et al., 2021). Whenever possible, all statistical estimates of population parameters will be tabulated along with corresponding confidence intervals (CIs) which will play a major role in data analysis. These analyses will allow us to examine the optimal timings on reaction times in the dot probe task for studying holding of attention.

There will be 6 unique conditions comprising 3 image pairs with 2 distinct timings each. Reaction times will be assessed for each of these conditions. Of particular significance are the two

image pairs that contain a face and a place, identified as the primary pairs given existing literature and our previous behavioral findings, suggesting the attention is captured and held by these stimuli. Conditions featuring both noisy squares serve as the baseline for comparison. It is hypothesized that reaction times will be lower (indicative of faster participant responses) when attention is captured and held to faces and places. Consequently, diverse display times will be used to determine variations in reaction times across conditions.

EEG Results

ERP waveforms will be grand averaged from lateral occipital sites (PO8 and PO7) for SPCN, N2pc, and P_D and the temporal ones (T5 and T6) for IIN. Separate waveforms will be plotted for ipsilateral and contralateral sites relative to the trial of interest (in this experiment, the face and place images). For example, the contralateral waveform for place images will be derived from the left-hemisphere electrode (PO7) when the place image is in the right visual field and the right-hemisphere electrode (PO8) when the place image is in the left visual field. The ipsilateral waveform will be the average of the right-hemisphere electrode (PO8) when the place image is in the right visual field and the left-hemisphere electrode (PO7) when the place image is in the left visual field.

Separate one-way ANOVAs will be conducted for each ERP component (SPCN, N2pc, P_D, and IIN), with trial type (place-noise, face-noise, valid, and invalid in 100 and 900 ms SOAs). By analyzing these components separately for each trial type and SOA condition, we aim to clarify the role of attention in these processes, from early attention capture (N2pc) to sustained attention hold (SPCN) and suppression of distractors (P_D), as well as disengagement and reorientation (IIN).

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We expect to observe the SPCN and N2pc components primarily in the place-noise conditions, with specific predictions for 900 ms SOA. The SPCN is predicted to appear at 320 ms in the 900 ms SOA condition but not in the 100 ms SOA, suggesting that SPCN is associated with sustained attention hold, which requires more time to manifest. The N2pc is expected to appear at around 200 ms, in the 900 ms SOA condition for the place-noise trial type, but not in the 100 ms (according to our previous behavioral results not showing any capture for place-noise conditions in that SOA), indicating that attention is being captured at longer SOAs for place-noise conditions. For the face-noise conditions, the N2pc is expected to appear in the 100 ms SOA condition, but it is not expected to be there for the 900 ms SOA, indicating attention capture in the short SOA window. For the face-noise conditions, we predict the P_D component to be more positive (indicating suppression of a salient distractor) at approximately 100 ms after stimulus onset in the contralateral sites, primarily in the 900 ms SOA condition but not in the 100 ms SOA.

A separate one-way ANOVA will compare IIN amplitudes between the valid and invalid trial types across both conditions (face-noise and place-noise), with a predicted significant effect of trial type. Follow-up t-tests will examine the difference between valid and invalid trials, with the expectation that IIN amplitudes will be more negative (indicating greater attentional disengagement) for invalid trials. Specifically, we expect to observe this component in both valid and invalid trial types for both conditions, but with a more negative amplitude (reflecting greater attentional reorienting) in invalid trials. The IIN is predicted to appear within the 220–280 ms time window at temporal sites (T5 and T6). This component is expected to be more pronounced in the invalid trials, where the target appears in an un-cued location, requiring a shift of attention away from the cued location to the new target. We hypothesize that invalid trials will show greater negative-going

IIN amplitudes, reflecting stronger attentional disengagement and reorientation compared to valid trials.

In summary, this experiment is novel in pairing EEG with a dot-probe task to identify neurophysiological markers of attention hold. The study is anticipated to demonstrate that attention hold is associated with the SPCN component, previously known as a working memory marker. While our prior behavioral results have shown that place-noise conditions elicit attention hold effects at longer SOAs, the ERP correlates of this behavioral outcome have not been thoroughly explored. This study will therefore provide an initial link between behavioral and neurophysiological measures of attention holding.

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